Jun Woo Beck

Audio & Music Machine Learning Researcher · Music Informatics Group, Georgia Tech

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RESEARCH SUMMARY

MS student in Music Technology at Georgia Tech's Music Informatics Group, researching machine learning and music information retrieval for sound effects and music audio. First author of a DAFx 2026 paper; planned thesis on multimodal music recommendation for video. Brings professional musicianship and audio engineering experience to audio ML research.

RESEARCH INTERESTS

Sound effects & music-audio machine learning · music information retrieval (MIR) · multimodal learning (audio, video, language) · audio dataset construction · interactive & biofeedback audio (secondary)

EDUCATION

Georgia Institute of Technology — Atlanta, GA
Master of Science in Music Technology — Aug 2025 – Expected May 2027
Music Informatics Group · Advisor: Dr. Alexander Lerch
Relevant coursework: Machine Learning, Audio Content Analysis, Integrating Music into Multimedia

Berklee College of Music — Boston, MA
Bachelor of Music in Electronic Production and Design — Aug 2024
Minors: Game and Interactive Media Scoring; Screen Scoring
Magna Cum Laude · GPA 3.71 / 4.00
Relevant coursework: Digital Signal Processing for Music Production and Postproduction, Music Acoustics

PUBLICATIONS

Peer-Reviewed Conference Proceedings

Beck, J. W., & Lerch, A. (2026). *Sound Effects Dataset Unification with the Universal Category System*. In Proc. International Conference on Digital Audio Effects (DAFx 2026), Boston, MA. [accepted; oral presentation, Sept 2026]

Oh, E. J., **Beck, J. W.**, & Smith, A. (2026). *The Singing Skin: An Audience-Centered Biofeedback System for Musical Interaction Based on Galvanic Skin Response*. In Proc. International Computer Music Conference (ICMC), Hamburg, Germany. [in press]

RESEARCH EXPERIENCE

Music Informatics Group, Georgia Institute of Technology — Atlanta, GA
Graduate Researcher — Advisor: Dr. Alexander Lerch — Aug 2025 – Present
Research spanning sound-effects dataset unification (**DAFx 2026**) and an industry-sponsored acoustic evaluation with **Textron**; thesis on multimodal music recommendation for video beginning Fall 2026.

SELECTED PROJECTS

Multimodal Music Recommendation for Video — Master's Thesis (planned) — From Fall 2026

- Thesis project to develop a multimodal model recommending fitting music for silent video by combining visual and text inputs to produce natural-language music recommendations.

Sound Effects Dataset Unification — EnvSound-UCS (DAFx 2026) Fall 2025 – Spring 2026

- Built a rule-based tag-to-UCS conversion framework reaching 98.5–100% automatic conversion across FSD50K, AudioSet, and ESC-50.
- Released EnvSound-UCS, a unified 58,057-clip environmental-sound corpus spanning 59 categories, and validated it with audio-classification benchmarks.
- Open-sourced the conversion pipeline, tools, and converted datasets across four public repositories (github.com/JunWooBeck).

Bio Sing-Along — Interactive Biosignal Performance Spring 2026

Premiered May 2026 at PRISM Pt. 1: Sound Through New Lens, Georgia Tech Center for Music Technology · with Eun Ji Oh

- Built the audio system — sound morphing and real-time processing — converting live GSR and PPG biosignals into the performance's sonic material.

Textron Panel Project — Acoustic Evaluation & Digital EQ Spring 2026

Industry collaboration with Textron · Music Informatics Group, Georgia Tech

- Conducted controlled acoustic measurements (THD, SINAD, frequency response) of speaker-panel audio quality in an isolation room against a consumer-laptop reference.
- Designed parametric EQ tuning presets that substantially reduced measured harmonic distortion and flattened the frequency response.

The Singing Skin — Audience-Centered Biofeedback System (ICMC 2026) Fall 2025

- Designed and implemented the full audio system translating real-time galvanic skin response (GSR) signals into interactive musical output.

HONORS & AWARDS

GTCMT Seed Grant ($2,000) — Georgia Tech Center for Music Technology 2026

Magna Cum Laude — Berklee College of Music 2024

Dean's List (4 semesters) — Berklee College of Music 2020 – 2023

SKILLS

Programming: Python, Max/MSP

Machine Learning: model training & evaluation; audio dataset construction & curation

Audio Engineering: recording & mixing (Pro Tools, Logic Pro, iZotope RX)

Sound Design: synthesis and sound-effects design for games, film, and video (FMOD, Wwise)

Music: composition, string arranging & orchestration; violin performance (classical & jazz)

MUSIC PRODUCTION & PERFORMANCE

Recording / Mixing / Live Sound Engineer — Seoul 2021

Mixed and mastered orchestral music with traditional Korean instruments (**Korean Ministry of Culture, Sports and Tourism**, dir. Echae Kang); recorded and mixed a full string-orchestra performance and designed live sound effects for dance (Seoul Ballet Festival, **Seoul Arts Center**).

Violinist & String Arranger — Dear Jazz Orchestra · Other Orchestras 2017 – 2022

Soloist and orchestra member at the **Seoul Jazz Festival and Jarasum Jazz Festival**; recorded on multiple Dear Jazz Orchestra releases and arranged strings for "Sing for You Vol. 1"; performed with Berklee World Strings, Berklee Motion Picture Orchestra, and film-scoring session orchestras.